

Review Article

Role of glycosylation in nucleating protein folding and stability

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Glycosylation constitutes one of the most common, ubiquitous and complex forms of post-translational modification. It commences with the synthesis of the protein and plays a significant role in deciding its folded state, oligomerization and thus its function. Recent studies have demonstrated that N-linked glycans help proteins to fold as the stability and folding kinetics are altered with the removal of the glycans from them. Several studies have shown that it alters not only the thermodynamic stability but also the structural features of the folded proteins modulating their interactions and functions. Their inhibition and perturbations have been implicated in diseases from diabetes to degenerative disorders. The intent of this review is to provide insight into the recent advancements in the general understanding on the aspect of glycosylation driven stability of proteins that is imperative to their function and finally their role in health and disease states.

Introduction

What is glycosylation?

Glycosylation is the covalent addition of sugar moieties to the protein co-translationally or as a post-translational modification enzymatically. It is the most frequent, heterogeneous, stereo-chemically complex modification. The rich repertoire of this modification serves as a profitable mechanism for augmenting the potential of the genome. A recent review throws light on the multifunctional role of glycans in organismal diversity [1].

A majority of the glycosylation sequons displayed in the protein are glycosylated and more than half of the proteome constitutes glycoproteins [2]. The site of modification is dependent on factors like the residues in the consensus sequence, its accessibility and the sub-cellular localization of the glycoprotein. During glycosylation, an oligosaccharide chain is most often linked covalently to either the nitrogen atom of an asparagine side chain or the oxygen atom of a serine or a threonine residue. Less common forms of glycosylation include C-glycosylation that occurs on tryptophan or tyrosine residue, S-linked glycosylation through a sulphur atom on cysteine or methionine and C-mannosylation through C–C linkages to the C2 position of tryptophan.

This formation of the amino acid and sugar linkage is an essential step in the glycosylation of the protein via a series of complex enzymatic reactions. These distinctive reactions are tissue-specific and have a significant impact on several physiological and pathological processes [3]. It is beneficial to the protein to undergo this modification during synthesis, and it has been noted that structural destabilization improved glycosylation efficiency while the natively folded substrate protein was not efficiently glycosylated *in vitro*. Initially, all the glycans are identical and undergo further processing via the trimming enzymes. This glycosylation code correlates with the biophysical effects of protein sequence and structure on glycosylation, and in this manner, it imparts the functional blueprint for the glycoprotein [4].

The endoplasmic reticulum quality control (ER-QC) recognizes aberrant proteins based on their glycosylation profile, targeting them for either rescue, export out of the ER or degradation. Modification of the glycan occurs as it proceeds into the Golgi apparatus, by the action of resident

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processing enzymes — glycosidases and glycosyltransferases forming an array of glycoforms. Once the protein attains the native conformation, the modified glycan serves as an address directing the protein trafficking towards endosome, lysosome, other compartments in the cell or by default towards the plasma membrane for secretion (Figure 1). In case of proteins with slow folding kinetics, they are targeted to the ER-associated degradation (ERAD) pathway for degradation by the mannose-trimming enzymes [5].

Despite not being under direct genetic control, glycosylation seems to be crucial in several aspects of the activity of the protein, that is its conformation, function, stability and dynamics. A fully processed glycan also has a huge impact on trafficking and localization of the glycoprotein and plays a major role in regulating the social behaviour of the cell including cell growth, differentiation and host cell interactions [6].

N-linked glycans have also been known to function as molecular insulators. They enable a reduction in the density of macromolecular crowded environment by retarding the intermolecular interactions and channelize the membrane proteins to function correctly. A recent study has demonstrated this effect on the diffusion of GPI-anchored variant surface glycoprotein (VSG) in the case of African trypanosomes. N-glycosylation was seen to modulate the molecular crowding threshold and facilitate lateral diffusion, high VSG mobility and dense packaging in their surface coat. It is hypothesized that the buffering effect offered by N-glycans can be crucial for antigenic variation leading to the evolution of new mosaic VSGs [7]. Studies have also established the role of a single glycan residue as being sufficient for receptor activity, protein trafficking and localization and bile acid uptake in the case of Na⁺-taurocholate co-transporting polypeptide, a receptor for hepatitis B virus. The perturbations in the glycan pattern led to protein degradation in the lysosomes via the endosomal pathway [8].

Additionally, individual glycosylation on a protein is known to have discrete roles in sorting, maturation, stability and function. A study was conducted to delineate the role of glycosylation site in the protein by sequentially removing and subsequently determining the effect on activity and fate of tyrosinase and tyrosinase-related proteins. It was seen that glycosylation on residue Asn 104 had no effect in protein processing or maturation, on Asn 94 it did not enhance stability or function while on residue Asn 304 (TRP-1), it was found to be important for activity and processing, and on Asn 181, glycosylation affected the glycoprotein transport through Golgi complex (Figure 2) [9,10]. A study conducted on similar lines on the human Ceroid-lipofuscinosis neuronal protein 5 based on the localization and stability assays revealed that the eight putative N-glycosylation sites of the protein could be categorized into three groups on the basis of the functionality it induced. In the first group, glycosylation on the sites Asn 179, Asn 252, Asn 304 and Asn 320 was found to be crucial in folding, failing which the protein was misfolded and was retained in the ER. The second group was involved in the trafficking and localization of the protein in the endosome/lysosome; here, Asn 401 glycosylation was found to be essential for lysosomal localization without which the protein was seen to accumulate in the Golgi initially before being transported to the plasma membrane for secretion. The third group was found to play a role in the lysosomal function where Asn 192 was found to be a crucial element for functionality [11]. These studies could figuratively draw lines to demarcate the functional differences in various N-glycosylation sites on the protein.

Types of glycosylation

A study on the nature and the distribution of protein glycosylation has revealed that the diversity of the glycans is achieved from the assembly of a total of 41 bonds between 13 different monosaccharides — glucose, mannose, galactose, fucose, N-acetylglucosamine, N-acetylgalactosamine, sialic acid, etc. and eight amino acids that are involved in glycosidic linkages [12,13].

These can further be classified into six types of glycosylation — N-linked glycosylation, O-linked glycosylation, C-linked mannosylation, S-linked glycosylation, phospho-glycation and glypiation. Among them, N- and O-linked glycosylations are the most common modifications seen in the glycoprotein (Figure 3) [14].

N-linked glycosylation

A vast majority of the secretory proteins, that is, ~90% of glycoproteins that occur in eukaryotes and archaea, undergo N-linked glycosylation in the ER. Here, upon recognition of the sequon N-X-S/T (or, rarely, N-X-C in a few plant proteins), where X can be any amino acid but proline, a 14-sugar glycan is transferred *en bloc* onto the amino group of asparagine during translation. On the other hand, not all the sequons in the proteins are glycosylated which can be due to varied reasons such as conformational limitations and amino acid environment. Data analysis has revealed that sequons containing proline and charged amino acids were significantly

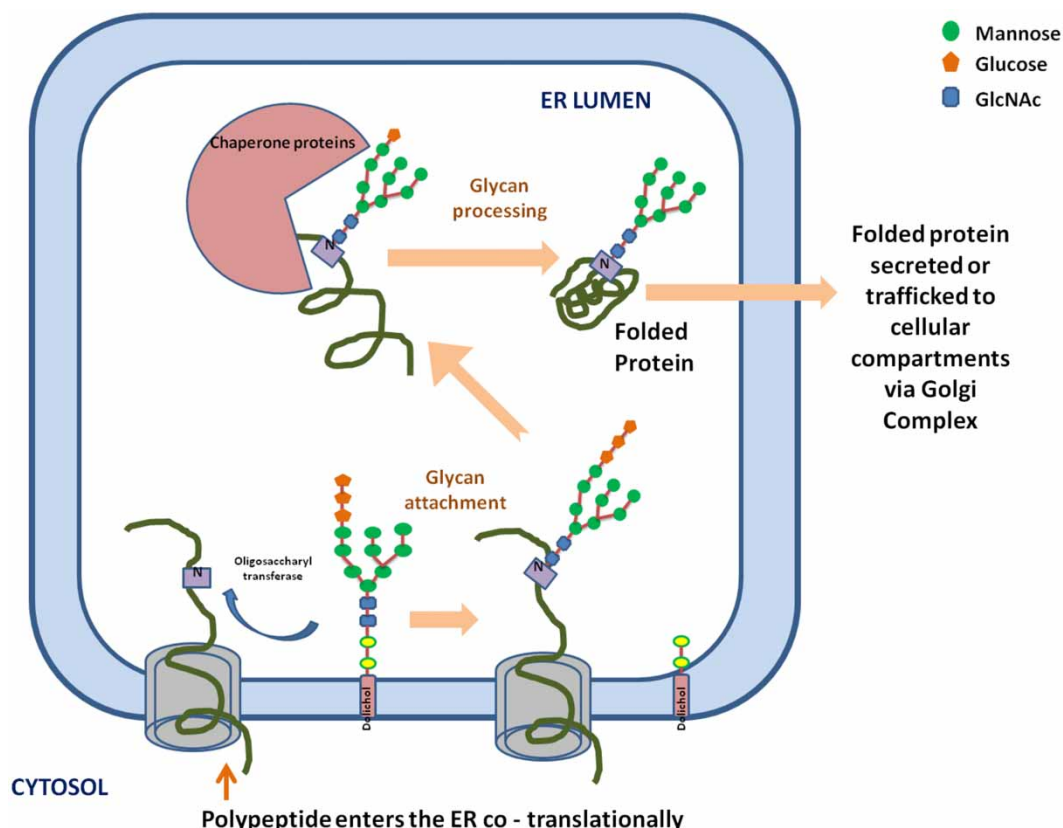


Figure 1. A schematic representation of the synthesis and processing of glycosylated protein in the ER.

The figure illustrates the glycan attachment onto the protein at the sequon N-X-T/S as it enters the ER. The folding of the protein is enabled by the lectin chaperones — calnexin or calreticulin. This is followed by glycan processing to target the glycoprotein to its destination. The folded protein is then secreted or trafficked to the other cellular compartments via the Golgi complex.

under-glycosylated compared with their uncharged counter parts. This results in micro-heterogeneity with a high degree of structurally complex glycoproteins [15].

Studies have further revealed that the presence of an aromatic residue at $n-2$ position of the sequon increases the glycan occupancy in human cells by enabling the activity of the enzyme oligosaccharyltransferase. This residue enables attainment of a reverse turn owing to the dispersion forces between the sequon residue and the aromatic amino acid [16]. It further alters the glycan processing in the Golgi complex producing less complex glycosylated structures. This demonstrates the sequence–structure–function correlation where a manipulation in the sequence space alters the structure and function of the glycoprotein [17].

O-linked glycosylation

O-glycosylation is known to be vital for the physiological function of proteins and plays a crucial role during the early stages of development by regulating cell growth and cell cycle. It serves as a mechanism to modulate protein–protein interactions and intercellular signalling [18–21]. Aberrations in O-glycosylation are implicated in certain human diseases and are also associated with disease risk factors [22]. The initiation event for glycosylation is mainly dependent on the localization of the glycosyltransferases involved in the transfer process [23]. Unlike N-linked glycosylation, sugars here are added in a step-wise manner catalyzed by different glycosyl transferase enzymes incrementally. Depending on the protein to be glycosylated and its localization, different glycans are attached to the proteins. The O-linked glycosylation occurs post-translationally and is initiated in the ER for most of the O-glycans where a single monosaccharide is linked to the hydroxyl group of serine, threonine, tyrosine or on the oxidized forms of lysine and proline via glycosidic linkages. These can be further extended in the Golgi compartment in the case of secreted and membrane proteins. O-linked glycosylation can

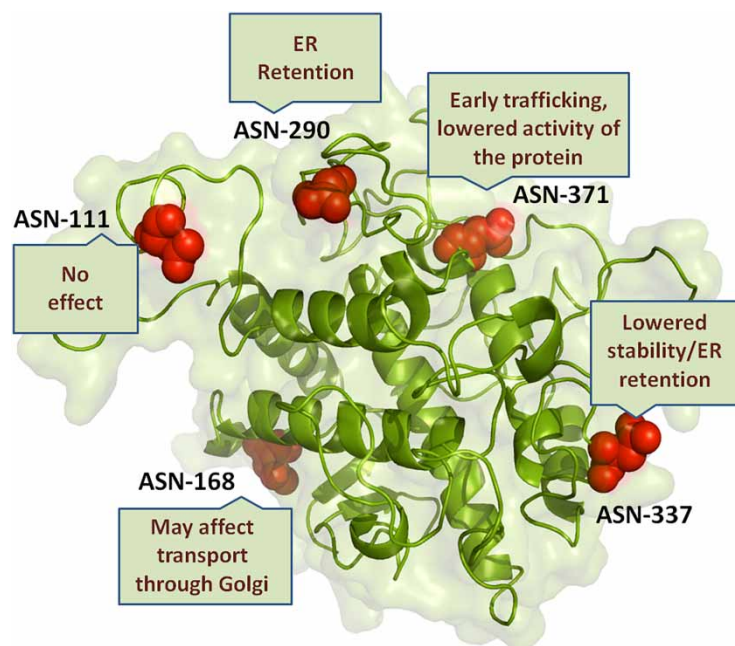


Figure 2. Graphical representation of the comparative roles of the conserved asparagine residues in human tyrosinase indicating the different functions they perform.

The human tyrosinase structure was modelled with tyrosinase from *Bacillus megaterium* as the template using the SWISS-MODEL Server. Glycosylated asparagine residues are shown in sphere representation.

be classified into O-glucosamidation, O-xylosylation, O-glucosylation, O-galactosylation, O-mannosylation and O-fucosylation. O-mannosylation in particular begins in the ER when a mannose sugar is transferred from dolichol monophosphate-activated mannose onto serine or threonine residues of secretory proteins [24]. O-mannosylation is significantly involved in ER protein quality control called as unfolded protein O-mannosylation where it inhibits the aggregation of the unfolded protein and promotes its entry into ERAD pathway [25].

A predominant form of O-linked glycosylation involves the addition of *N*-acetylgalactosamine (O-GalNAc) onto the serine or threonine side chain catalyzed by the enzyme *N*-acetyl galactosaminyltransferase in the Golgi apparatus. Mucin proteins, extracellular matrix components and proteoglycans frequently undergo this form of glycosylation [26]. Glycosylation of hydroxylysine and hydroxyproline is initiated in the ER where a Gal or a Glc residue is added onto collagen proteins which can be extended further in the Golgi complex. On the other hand, proteins of the nucleus and cytoplasm undergo O-GlcNAcylation where a serine/threonine residue is reversibly glycosylated with GlcNAc. This modification is intimately associated with cell signalling where it adds an additional level of modulation over and above what is done by protein phosphorylation alone [27].

To understand the effect of glycosylation on protein folding, stability and dynamics, it is imperative to study the glycosylation code and its influence on the protein's energy landscape.

Effects of glycosylation on protein stability

The effect of glycosylation on protein folding is of great interest as it has a tremendous impact on the stability and structure of the protein. The covalent bonding of glycans to proteins enhances their thermal and kinetic stability by increasing their vibrational entropy, which forms a major portion of the configurational entropy in determining the protein dynamics and decreasing the free energy of their folding. It stabilizes the pre-existing conformation, reduces its tendency to form other secondary structures and prevents thermal unfolding. These auxiliary effects arise from multiple factors, including the presence of a disulphide bond adjacent to the glycosylation site which often promotes the development of secondary structures.

Earlier experimental and *in silico* studies have shown that glycosidic linkages on the Asn residue induce local stabilization effect around the glycosylation site [28]. An analysis of the heptasaccharide glycan present in

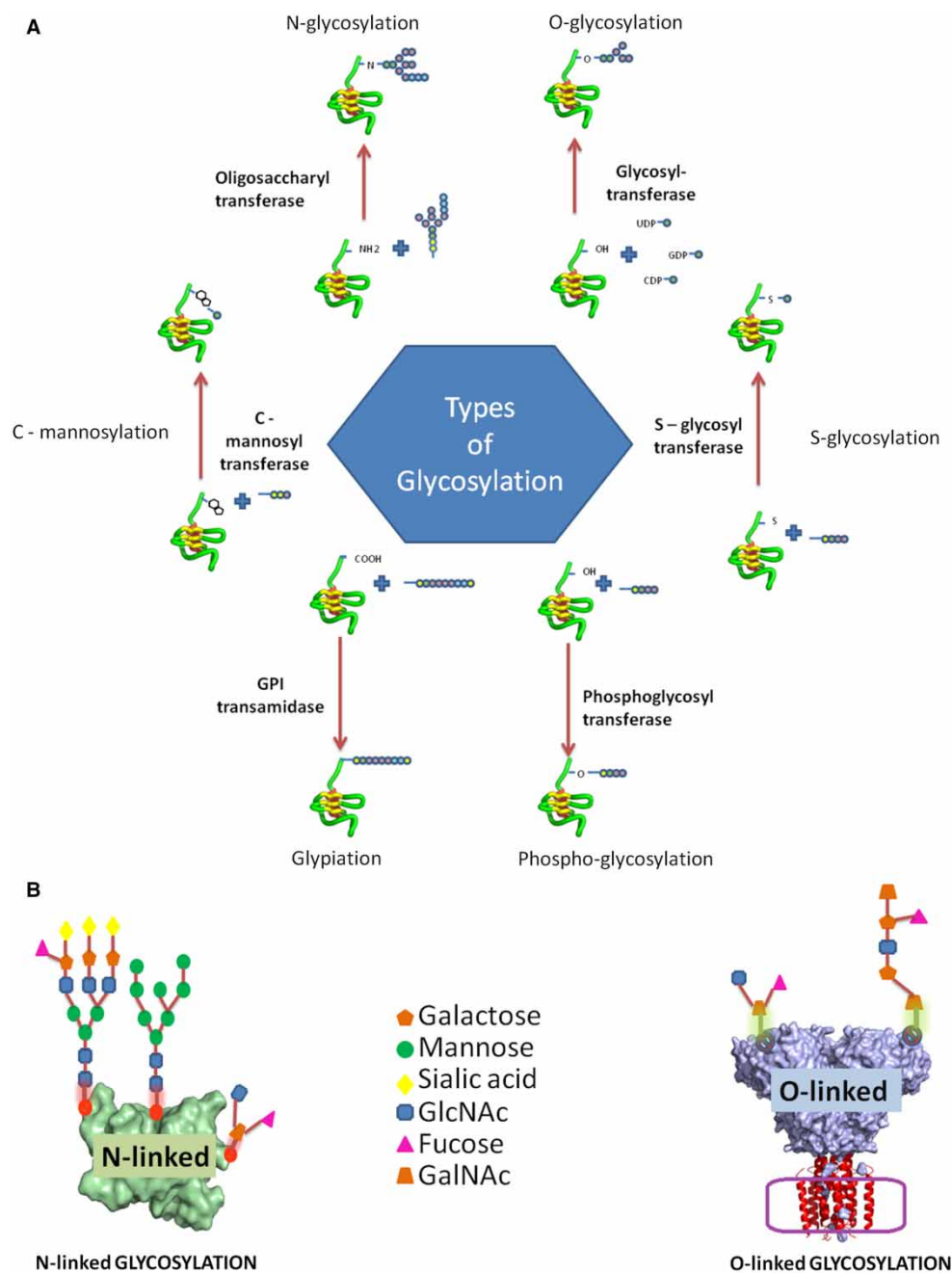


Figure 3. Types of glycosylation.

(A) A graphic representation of the six types of glycosylation, their site of glycan attachment and the various glycosylated proteins. (B) Proteins covalently linked to N-linked and O-linked glycans that are differentially processed.

ECoRL (lectin from *Erythrina corallodendron*) has shown that the presence of the glycan guides their folding pathway probably by stabilizing certain secondary structural elements of intermediates during folding [29]. The solution and molecular dynamics studies have advanced the study of the conformation of the oligosaccharide and its effect on the structure and stability of the glycosylated and the non-glycosylated recombinant rECoRL. The study indicates that, although the three-dimensional structures of the two forms are alike, the glycosylated form displayed a reduction in the non-polar solvent accessible surface area when compared with the non-glycosylated ECoRL protein. These findings provide a rationale for the higher thermodynamic stability of the glycosylated protein in comparison with its non-glycosylated counterpart [30].

A similar assistance of glycans in the folding process was seen for another glycoprotein, soybean agglutinin (SBA), where glycosylation had a pronounced effect in maintaining the conformational stability of the lectin [31,32]. A study on the structural perturbations during the thermal unfolding of the two forms — glycosylated and non-glycosylated forms of SBA — shows that the non-glycosylated monomeric form has lesser stability than its corresponding glycosylated form at normal and elevated temperatures from the dynamics perspective. This is attributed to the non-covalent interactions between the protein and the carbohydrate as seen in the glycosylated protein. The results from Molecular dynamics simulation studies correlates the decrease in free energy, with the increase in vibrational energy as a result of glycan-induced stabilization by ~45 kcal/mol at normal temperature. Studies also revealed that the non-glycosylated monomeric form of the legume lectin SBA was less stable than the corresponding glycosylated form at normal and elevated temperatures. This was seen in terms of the interaction profile between the carbohydrate and the protein which was apparently crucial for maintaining the folded structure of the legume lectin (Figure 4) [33,34].

Studies further highlight the role of glycosylation in terms of the protein–glycan interactions. Coarse-grained molecular dynamics simulations conducted to understand the mechanism underlying the loss of stability of the MM1 protein (miniaturized model of Gc-macrophage activating factor) by glycosylation, point to the origin of the destabilization to a conformational deformation of the protein led by the interaction of the monosaccharide with the protein surface. Loss of long-range protein–protein interaction has a greater destabilizing effect than the creation of new short-range protein–glycan interaction. This originates from a trade-off between the short-range and long-range interactions [35].

A study on the effect of glycosylation on the dynamics of the Ribonuclease B protein revealed that although N-glycosylation did not significantly alter the global or the local structure, it did reduce the dynamic fluctuations, promoting an increase in stability. The global structures of glycosylated proteins are almost identical with non-glycosylated ones negating the effect of N-glycosylation on the global protein structure. However, the data from the NMR measurement of amide-proton/deuterium exchange rates indicated that the glycosylation of the Ribonuclease B protein enabled the protection of amide-proton resonances from solvent exchange for residues both around and far away from the glycosylation site. This was evidently not seen in the non-glycosylated form. This result further indicates that the presence of sugar enhances the protein stability [36].

Studies also show the influence of glycosylation in maintaining the biological activity of the protein. Glycodelin, a human secreted glycoprotein, for example has been of considerable interest to the understanding of the influence of glycosylation on its biological and the physiological activity in human reproduction. The N-linked glycans on glycodelins differ between the male and female glycoforms. Glycans on the glycodelin proteins are believed to assist the pathogens and infected cells to escape host immune reactions [37]. It has also been reported that insignificant elaboration of glycans on the Fc portion of IgG can have a significant impact on the receptor binding and manipulating effector functions [38,39]. A similar study on the sphingomyelin phosphodiesterase acid-like 3A protein also established the various roles of N-glycosylation in preserving its phosphodiesterase activity, intracellular stability and protecting it against the degradation enzymes. The study further experimentally verified that the cleavage of the glycan from a matured protein resulted in loss of its enzymatic activity. It was also seen that the non-glycosylated protein was misfolded and was subjected to rapid degradation [40].

O-glycosylation that is mannosylation in particular has been shown to indirectly promote stability in ER-associated secretory proteins Sec20p by hindering proteolytic cleavage [41]. It is also known to increase the thermostability of carbohydrate-binding modules by increasing the thermolysin cleavage resistance along with the binding affinity towards its substrate — bacterial microcrystalline cellulose [42]. There are, however, cases wherein no significant differences in terms of activity, conformation and dynamics between the glycosylated and the non-glycosylated forms were observed. A recent study on the Artemin (ART) protein revealed that the glycosylation offered no suggestive role in the activity of the protein despite the glycosylation sequon being at a close proximity to the protein active site. There was, however, a clear role played by glycosylation in increasing the solubility and stability of the protein [43]. These studies emphasize that natural glycosylation results in protein stabilization.

Effect of glycosylation on protein folding

Glycosylation exerts its intrinsic effect by directly transforming the energy landscape for protein folding. It induces a locally mediated effect on the protein structure. These effects vary in accordance with the glycosylation site and its secondary structure. It is known to negatively affect the α -helical segment if the glycosylation

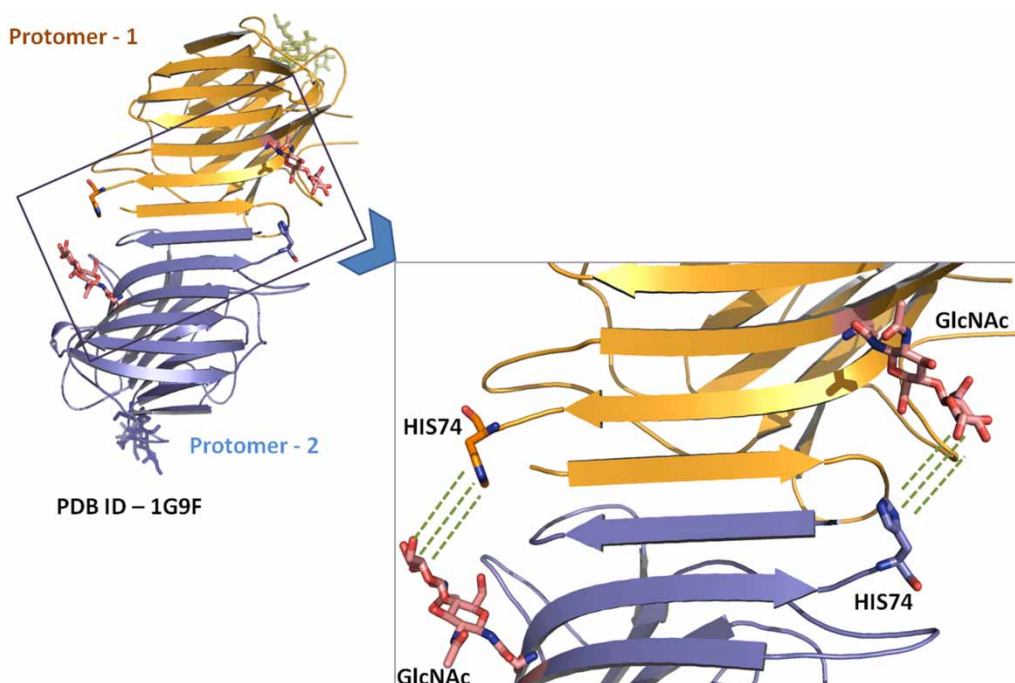


Figure 4. Cartoon representation of non-covalent interactions that stabilize the functional form of soybean agglutinin. The non-canonical interface is highlighted in the box. The interaction between the two dimers is glycan modulated and is indicated by dashed lines.

site is at the centre of the helix while positively promotes folding if the site is between the ordered secondary structures. Previous studies have established that the three residues closest to the protein play a major role in accelerating protein folding and enable stabilization. This endorses the reasoning behind the most preferred section of the protein to get glycosylated. Studies have also shown the consequence of specific, evolved protein–glycan contacts in mediating the beneficial energetic effects on protein folding that is glycosylation facilitated [44–46].

Glycosylation is seen to transform the folding kinetics of proteins by altering the rate-limiting step of the protein folding reaction. In the case of erythropoietin, it was observed that the absence of glycosylation did not alter the folding mechanism but stabilized the intermediate species to a greater extent than increasing the erythropoietin equilibrium stability. These results suggest that by biasing particular folding pathways, glycosylation impedes the folding of the intermediates rather than accelerating them. It further increases the folding rate of the protein than optimization of the biological function than [47]. The findings from another study on the non-structural protein 4B (NS4B) from Dengue virus indicated that disruption of the glycosylation rendered loss in the biological function of the protein in the formation of viral replication complex by adversely affecting its interaction with other proteins in the replication complex. Mutational analysis revealed the role of glycans in maintaining the folded conformation of the NS4B protein [48].

On the other hand, a study on the role of fucosylation in the structural stability and function of PMP-C (pars intercerebralis major peptide-C) did not display any striking structural changes owing to fucosylation in its native disulphide-bonded forms of PMP-C. It did, however, address the means of enhancing the function of PMP-C as a protease inhibitor via the presence of disulphide bonds between conserved cysteines and fucosylation, thereby preserving the structural integrity [49]. Investigation of the structure, dynamics and stability of the PMP-C protein by NMR spectroscopy also acknowledged the fact that the fucosylated and defucosylated forms were very similar in their overall backbone conformations and were indistinguishable. The L-fucose moiety induced a local structural effect largely restricted to the side chains of neighbouring residues inducing the formation of a hydrophobic cluster. Likewise, analysis of the exchange rates of amide protons asserted the

role of fucosylation in decreasing the dynamics of the protein which, in turn, led to an increase in stability as was evident by thermal denaturation studies [50].

O-glycosylation also has a crucial role in determining the structure of the protein in terms of the secondary, tertiary and quaternary structure of the folded protein. An example is the case of the human granulocyte macrophage colony-stimulating factor protein where O-glycosylation is seen to be essential for maintaining the structural stability of the protein and has resulted in an extended *in vivo* half-life without affecting its biological function. Indeed, its deglycosylation was seen to affect significantly the attainment of secondary structures particularly the α -helical ones [51]. In the context of intrinsically disordered proteins (IDPs), a simulation and dynamics study on the structural effect of O-linked glycosylation on IDPs revealed that O-linked glycosylation primarily modulates the functional property by altering the binding characteristics of the protein. Structurally, it was found to induce a reduction in the secondary structure propensity [52].

Molecular mechanism by which glycosylation helps in protein folding

Both N- and O-glycans serve as molecular appendages that stabilize the protein structures by the addition of bulky hydrophilic groups inducing favourable interactions around the modification site. The rationale underlying this decrease in dynamics can be attributed to several mechanisms. For instance, the addition of a sugar moiety can provide a unique conformational change to the inter-glycosidic and glycopeptides linkages which dictate a precise conformation for the peptide reducing its dynamics considerably [53]. Core glycans enhance conformational stability thereby productive folding of the proteins [54]. A related study suggested two possible microscopic mechanisms of glycosylation on protein's stability which is termed as a 'context dependent' glycosylation effect; the context here is the protein's interaction network around the glycosylation site one where it energetically stabilizes specific hydrogen bonds by shielding them from solvent exposure leading to changes in the hydrophobicity of the modification site which is experimentally verified [55,56]. The second mechanism being entropic stabilization which affects local and large-scale correlated motions of protein [57]. Furthermore, studies on the effect of glycosylation on protein conformation have revealed that glycosylation facilitates structural integrity of the protein by increasing the favourable interactions like hydrogen bonds, van der Waals, electrostatic energy between the protein and the glycan components enabling protein folding. This was observed in 2384 N-glycoproteins retrieved from the Protein Data Bank (Figure 5) [58,59].

Thermodynamic studies on the human CD2 (cluster of differentiation 2), a cell surface glycoprotein, aimed to provide quantitative energetic analysis on the effect of N-glycosylation showing that glycosylated hCD2 is more stable by 3.1 kcal/mol, with the folding accelerated by 4-fold, and has a significantly slower unfolding rate (50 times) than its corresponding non-glycosylated form. This difference was attributed to the three hydrogen-bonding interactions within the type I β -bulge turn providing the required energetic contribution for the protein stability [60]. It was also seen that the N-glycan's first saccharide unit was responsible for the entire acceleration of folding along with 75% of the native state stability. The other two saccharide units contributed in achieving the remaining 25% of the stability that was required. The study thus illustrated the role of the conserved N-linked triose core, ManGlcNAc(2), in improving both thermodynamics and kinetics of protein folding [45].

Coarse-grained MD studies have shown that introduction of glycans of a suitable hydrophobicity at flexible regions aided folding of the protein and increased the conformational stability in comparison with the native counterpart. However, the same was not true in the case of glycan addition onto the restricted residues as their contribution to stability was limited but did increase the free energy barriers significantly. The glycosylation was found to follow the two-step folding mechanism that is hydrophobic collapse followed by structural rearrangement. Glycan chains, which were appropriately located in terms of hydrophobicity, promoted both the collapse and rearrangement while others were found to only accelerate the collapse hindering the structural rearrangement. Complementary to the entropic effects, there is a reduction in the conformational space of the unfolded state wherein suitably located glycans provide the required hydrophobic interactions strengthening the folded state [61].

These studies highlight the role of glycosylation in reducing the overall dynamic fluctuations of the protein, thereby accelerating protein folding and maintaining the folded protein conformation. The general mechanism for protein folding is thus a result of the decrease in the activation energy which induces native-state stabilization and enhances protein folding in the secretory pathway.

Thermodynamic and kinetic aspects of glycosylation

The covalent binding of the bulky hydrophilic glycans is known to increase the solubility of the protein enhancing the thermodynamic and kinetic stability of the glycoprotein, as well as providing protection from

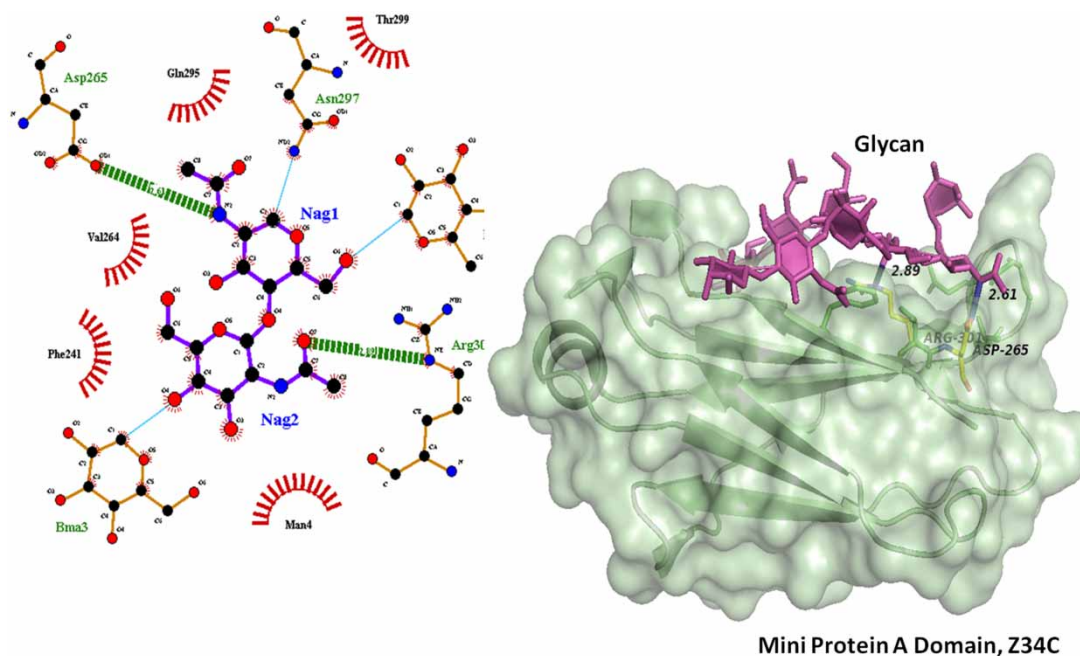


Figure 5. Interaction plot illustrating the molecular mechanism by which glycosylation stabilizes the protein structure. Hydrogen bonds and hydrophobic interactions are depicted in the 2D plot. This was generated using LigPlot+. The protein and glycan are shown in the 3D plot by surface and stick representation, respectively. Hydrogen bonds between the glycan and the protein, which are crucial for the interaction are indicated in blue.

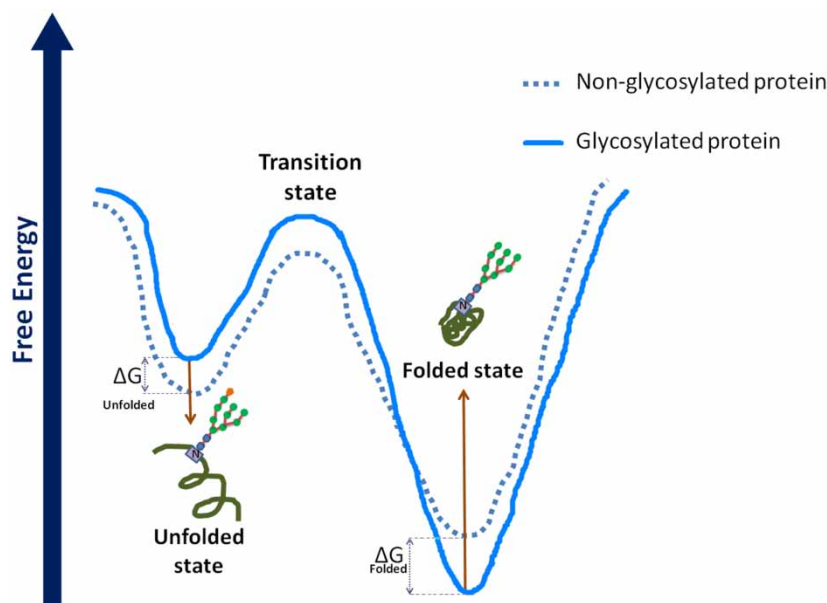


Figure 6. A schematic representation of the free energy profile of a glycoprotein.

Glycosylation is known to result in higher stability of the glycosylated form than the non-glycosylated form in their native state. It is also known to affect the height of the free energy folding barrier of the protein. Image adapted from Shental-Bechor and Levy [65].

proteolysis. This increase in thermodynamic stability is manifested by the increase in the melting temperature along with the free energy difference between the two forms. Studies also report enhanced stability in case of N-linked glycoproteins which is not the case for O-linked glycosylation [62–64]. The extent of change in stability is, however, glycoprotein-dependent and is essential to decipher the glycosylation code that directs the interaction between the nature of the carbohydrates or the proteins, thereby dictating the biophysical properties of the glycosylated protein [4]. Thermodynamic analysis to investigate the origin of the enhanced protein stabilization by glycosylation indicates that the difference mainly arises from the higher amount of destabilization of the unfolded state in comparison with stability gain of the folded state. This was observed in the case of 63 engineered SH3 domain variants with varied polysaccharide chains attached to glycosylation sites, wherein the thermal stabilization of the protein was found to be proportionate to the number of covalently attached glycans along with the position of the glycosylation sites. The analysis indicated that the enhanced protein stabilization originated by destabilization of the unfolded state reasonably than stabilization of the folded state. Conjointly, it was also observed that the bulky polysaccharide chains forces the unfolded ensemble to adopt a more extended conformation. This is mediated by limiting the formation of a residual structure besides enthalpically inducing a higher free energy to the unfolded state (Figure 6) [65].

Studies on the kinetics of thermally induced aggregation of the glycoprotein *Peniophora lycii* phytase (Phy) and a deglycosylated form (dgPhy) discovered that the glycans strongly promoted kinetic stability by reducing the rate of irreversible denaturation while leaving the temperature for equilibrium denaturation unaltered. The deglycosylated form was found to aggregate ~200 times faster than the glycosylated counterpart. This suggested first-order kinetics with respect to the concentration of the thermally denatured state. Here, in the case of glycosylated form, the inhibition of aggregation was found to be dependent on steric hindrance of the glycans in the aggregated form of the protein and was not due to strong hydration of the glycans which was affecting their kinetic stability [66].

Investigation of the glycosylation patterns in the variable domain of the HIV1 Envelope protein (gp120) revealed that glycosylation reduced the formation of secondary structures in the peptide by stabilizing the pre-existing conformation. This effect is attributed to the increase in the entropy of the backbone, disruption of inter- and intra-hydrogen bonding interaction between the glycan and the peptide simultaneously resisting unfolding. Glycan–glycan interactions provided further stability by contributing enthalpically aided by di-sulphide linkages promoting the formation of β -strands. The study established the influence glycosylation exerted on the antigenicity and the immunogenicity of the variable region [67].

Furthermore, the oligosaccharide functions as a crowding agent. This results in a more expanded unfolded state with loss of residual structure and thus higher enthalpy. It has also been indicated that it was almost impossible to attain a higher stability for a protein structure than what was induced by the sugars [68].

Effect of glycosylation on protein oligomerization

Glycosylation has been seen to hinder oligomerization when the sequon is at the interface of the oligomer formation. A study on the effect of glycosylation on the dimerization of the human carcino embryonic antigen-related cell adhesion molecule 1 revealed that the native monomer of the molecule suggested extensive glycosylation at the interface. However, it was seen that addition of even a single N-linked N-acetylglucosamine at any of the potential glycosylation sites inhibited the dimer formation [69].

In the case of ECorL, atomic resolution structural solution comparison of the glycosylated and the non-glycosylated form revealed the presence of local changes near the glycosylation sites. These served as compensatory interactions for the non-glycosylated protein replacing the hydrogen bonds between the protein and glycan. Additionally, examination of the dimer discovered minor changes in the conformation of the residues in the interface. However, the final oligomeric structure of the ECorL was independent of the glycan attached to it as the tertiary and quaternary structures of both glycosylated and the non-glycosylated forms were equivalent [70]. A similar result was also seen in basic winged bean agglutinin (WBA) lectins wherein the dimerization was independent of the glycosylation as the site for the addition of sugar moieties was located away from the dimerization interface. It was further seen that in WBAIL, glycosylation reduced the structural integrity by reducing the number of inter-subunit interactions in the protein, lowering its denaturation temperature while ensuring a two-state thermal transition [71–73].

The study on two Ribonuclease (RNase) proteins A and B revealed that, although identical in sequence and conformation, the ability of oligomerization of protein RNase B is slightly hindered during denaturation exclusively. This was because of glycosylation in the former one at site N34 when compared with RNase A which

forms both N and C dimers via swapping of the respective termini of the protein. Although alike in conformation, the folding rates of the two proteins are faster in the case of the glycosylated form. The oligomerization study suggested that the oligosaccharide chain either favours productive collisions between the monomers or stabilizes the N-termini-mediated dimers by increasing the compaction of the oligomer (Figure 7) [74,75].

Prevention of oligomerization owing to glycosylation is seen to be beneficial in the case of Alzheimer's disease as the addition of O-GlcNAc was seen to hinder the progression of neurodegeneration by stabilizing both tau and TAK-1 proteins by inhibiting them against aggregation. In the case of tau protein, it was further seen that O-GlcNAc was competing with the sites of phosphorylation and in turn blocking hyper-phosphorylation and oligomerization. This study also identified O-GlcNAcase, a glycoside hydrolase to be a potential therapeutic target with the ability to alter the Alzheimer's disease progression [76]. These studies demonstrate that the mode of dimerization of these glycoproteins is not dependent on their glycosylation but is overseen by the features that are intrinsic to the protein.

Effect of impaired glycosylation in disease states

N-glycosylation of proteins has been implicated in several diseases as it is intimately linked to the early protein folding events. Incomplete glycosylation, prevention of the addition of sugar moieties or lack of a full N-glycosylation site occupancy leads to abnormal protein folding and alterations in the physical properties of the protein. This results in unfolded protein response (UPR) triggered due to ER stress response [77]. There

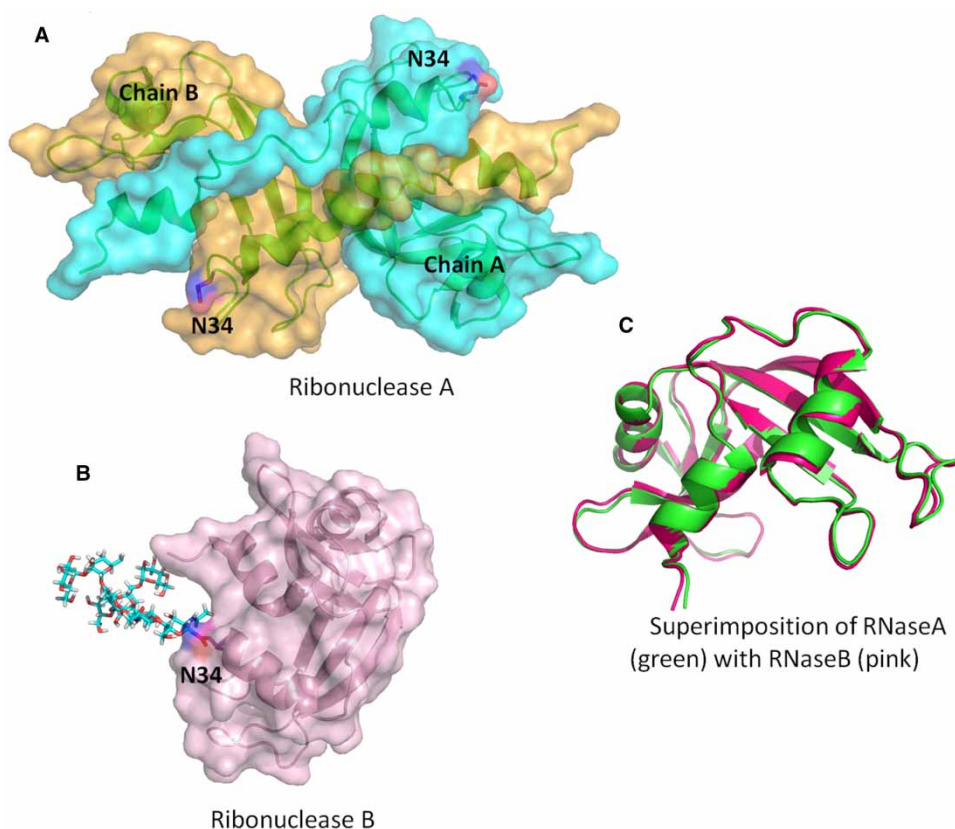


Figure 7. Oligomerization of the ribonuclease proteins.

(A) N-dimer of the Ribonuclease A protein. Chain A is coloured in cyan and chain B in orange. The putative glycosylation site which is devoid of the N-glycan is indicated in stick representation. (B) Monomer of the Ribonuclease B protein. Chain A is coloured in pink and the glycosylation site N34 is shown in stick representation covalently linked to the glycan Man₆GlcNAc₂. (C) Superimposition of the 3D structure of RNase A shown in green with RNase B shown in pink. The RMSD was found to be 0.475 Å.

are several known genetic diseases listed in the National Organization of Rare Diseases (U.S.A.) that arise either from improper assembly or processing of glycans which are collectively known as congenital disorders of glycosylation [78]. Studies have also established the effects of the *N*-glycan precursor length and their effects on protein folding [79].

Studies on Alzheimer's disease have revealed that improper glycosylation of the amyloid precursor protein leads to the accumulation of the misfolded protein. This accumulation elicits the UPR, which triggers aberrations in cellular responses such as autophagy and apoptosis as a prelude to Alzheimer's disease [80]. Furthermore, in the case of congenital muscular dystrophy, proper glycosylation of the dystroglycan protein along with O-mannosylation is crucial for its binding to the extracellular matrix necessary for maintaining the stability of the membrane. Unlike other glycosylation disorders, this protein reaches its destination, but is non-functional owing to improper conformation caused due to impaired glycosylation [81].

Mutations in the *cystic fibrosis* transmembrane conductance regulator (CFTR) protein have been implicated in cystic fibrosis (CF), one of the lethal autosomal recessive disorders. The most common mutation results in defective glycosylation of CFTR, thereby affecting the secretory and endocytic traffic leading to mislocalization. *N*-Glycans also alter the folding energetic of the channel. This altered glycosylation is believed to contribute to the chronic infection and continuous inflammatory assault in the CF lung [82,83]. Epithelial cadherin (E-cadherin), a calcium-dependent cell–cell adhesion molecule, has been known to be crucial in epithelial tissue morphogenesis. Alteration in the glycosylation profile of this protein promoting malignant transformation has implications for its function, that is change in the protein conformation in turn modulates cell–cell adhesion — a key pathological event in cancer metastasis [84].

In the case of voltage-gated calcium channels, alterations in the glycosylation pattern of the channel protein result in pathological conditions. This is mainly due to modifications in the expression and hence regulating the gating property of the channel. These channels have been identified to contribute to peripheral neuropathy associated with diabetes. Defects in glycosylation pattern of these channels are also seen in congenital disorders such as long QT syndrome and varitint-waddler phenotype, a mutant phenotype [85]. On similar lines, transient receptor potential channel (TRPP2) membrane protein is heavily glycosylated, is crucial for the biogenesis and stability of the protein and mutations at the glycosylation sites have been shown to affect the protein expression at the plasma membrane, highlighting its role in protein trafficking leading to autosomal dominant polycystic kidney disease [86].

In summary, impaired glycosylation has been concomitantly linked with misfolded protein which in turn affects its localization, the downstream functioning of the protein and is manifested in a disease condition, substantiating the role of glycosylation in the proteostasis.

Conclusion

Recent studies have established the role of glycosylation in a myriad of biological processes like the attainment of protein structure, oligomerization, protein trafficking, reduction in protein aggregation, modulation of protein function, ER-QC and in protein homeostasis. They mainly affect properties, such as protein solubility, polarity and bioavailability, and shield hydrophobic patches and sensitivity to protease enzymes. Studies have also explored the various cellular functions wherein the glycans play a critical role in cellular growth, differentiation, cell–cell communication, signal transduction, inflammatory and immune response. The impact of glycosylation for a protein is dependent on the location, chemical environment and number of glycosylated sites in the protein while on the other hand, the size, flexibility, structure and number of branches affect the glycosylation from the glycan's perspective. The consequence of glycosylation is seen by an increase in solubility enhancing the thermal and kinetic stability accompanied by an augmented vibrational energy and a decrease in the free energy of folding. Studies have further noted the accretion in structural integrity via the non-covalent interactions between the protein and the glycan. This network of interactions induces local stabilization around the glycosylation site. This further corresponds to the decrease in protein dynamics and thereby aids in protein folding and conformational stability. It is therefore imperative to understand the influence of glycosylation in terms of the folding energy landscape of the protein. Furthermore, the glycan was seen to promote productive collisions favouring the protein oligomerization. Additionally, specific changes in the glycosylation of the glycoprotein have been implicated in disease progression, and these serve as potential biomarkers for disease diagnosis, prognosis and prediction of treatments.

Despite these promising findings, we have barely begun to decipher the role of glycosylation on protein structure, dynamics and activity. Thus, it remains to be an area of intense scientific scrutiny.

Abbreviations

CD2, cluster of differentiation 2; CF, cystic fibrosis; CFTR, *cystic fibrosis* transmembrane conductance regulator; ECorL, lectin from *Erythrina corallodendron*; ERAD, ER-associated degradation; ER-QC, endoplasmic reticulum quality control; GPI, glyco phosphatidyl inositol; IDP, intrinsically disordered proteins; MD, molecular dynamics; NS4B, non-structural protein 4B; PMP-C, pars intercerebralis major peptide-C; RNase, ribonuclease; SBA, soybean agglutinin; UPR, unfolded protein response; VSG, variant surface glycoprotein; WBA, winged bean agglutinin.

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Competing Interests

The Authors declare that there are no competing interests associated with the manuscript.

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